The empirical recurrence for OEIS A184023, proved by splitting on the common sum

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Abstract

OEIS A184023 counts arrays over $\{0,\ldots,3\}$ in which all 2×2 subblock sums are EQUAL — the common value not being named — and carries an empirical recurrence of order 15 contributed by R. H. Hardin. The unnamed value is what stops a transfer matrix from applying directly, and it costs nothing to remove: an array with at least one 2×2 subblock determines its own common sum, so the count splits as a sum over the $12+1$ possible values with nothing counted twice, and each summand is an ordinary walk count. On the block-diagonal matrix over those values the recurrence is decided by a finite exact computation on $S = 3328$ vertices.

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1 The sequence and the conjecture

OEIS A184023 is "1/8 the number of (n+1)X4 0..3 arrays with all 2X2 subblock sums the same.". It has offset 1 and begins

$$242, 491, 1052, 2597, 6650, 18701, 53948, 165929, \ \ldots$$

The entry carries the following comment:

> Empirical: a(n)=8*a(n-1)+16*a(n-2)-268*a(n-3)+189*a(n-4)+3388*a(n-5)-5824*a(n-6)-20048*a(n-7)+49756*a(n-8)+52752*a(n-9)-196896*a(n-10)-24192*a(n-11)+369216*a(n-12)-131328*a(n-13)-262656*a(n-14)+165888*a(n-15)

As of the "Last modified" line on the live entry (Oct 03 2025 02:15:16, revision 9) the statement is still recorded as empirical, and nothing on the entry records it as settled either way.

Written out, the claim is

$$a(n) \;=\; 8\,a(n-1)+16\,a(n-2)-268\,a(n-3)+189\,a(n-4)+3388\,a(n-5)-5824\,a(n-6)-20048\,a(n-7)+49756\,a(n \tag{1}$$

for all $n > 15$.

2 Splitting on the common sum

The arrays have 4 columns and $L = n+1$ rows, entries in $\{0,\ldots,3\}$, and every 2×2 subblock has the same sum. Since $L \geq 2$ and $4 \geq 2$, every such array contains at least one 2×2 subblock, and that subblock's sum is the common value; so each array belongs to exactly one of the classes

$$A_c = \{\text{arrays all of whose } 2\times 2 \text{ subblock sums equal } c\}, \qquad 0 \leq c \leq 12,$$

and $\#\{\text{admissible arrays}\} = \sum_c |A_c|$ with no array counted twice. The bound $12 = 4 \cdot 3$ is the largest possible subblock sum.

For a fixed c the condition is local, so $|A_c|$ is a walk count. Let $V = \{0, \ldots, 3\}^4$ be the possible rows and let M_c be the adjacency matrix of the digraph in which $r \to s$ when $r_j + r_{j+1} + s_j + s_{j+1} = c$ for every j with $1 \le j \le 4-1$. Then $|A_c| = \mathbf{1}^\top M_c^{L-1}\mathbf{1}$, since an array of L rows is a sequence of L elements of V and each of its 2×2 subblocks lies in two consecutive rows and two consecutive columns.

Writing $N = \bigoplus_{c=0}^{12} M_c$, of size $S = 3328$, and $\mathbf{v}$ for the all-ones vector on it,

$$a(n) \;=\; \tfrac{1}{8}\,\mathbf{v}^\top N^{\,L-1}\mathbf{1}, \qquad L = n+1. \tag{2}$$

3 The criterion, and the computation

Let $q(t) = t^{15} - \sum_i c_i\, t^{15-i}$ be the characteristic polynomial of (1) and $G = N$.

Lemma 1. *With n_0 the least index for which $L \ge 1$, $o = 1n_0 + 1 - 1$ and $h_j = G^j N^o \mathbf{1}$, we have for $j \ge 15$*

$$a(n_0 + j) - \sum_i c_i\, a(n_0 + j - i) = \tfrac{1}{8}\,\mathbf{v}^\top G^{\,j-15} w, \qquad w = h_{15} - \sum_i c_i h_{15-i}.$$

Hence (1) holds from that point on if and only if $\mathbf{v}^\top G^m w = 0$ for all $m \ge 0$, and $m < S$ suffices.

Proof. Substitute (2) and factor. By Cayley–Hamilton G^S is an integer combination of $I, G, \ldots, G^{S-1}$, so every later value repeats that combination of earlier ones. □

Theorem 2. *The recurrence (1) holds for every $n > 15$, which is the statement of Section 1.*

Proof. Each M_c was built directly from its defining condition. The vector w was formed by 15 applications of G in exact integer arithmetic and $\mathbf{v}^\top G^m w$ evaluated for $m = 0, 1, \ldots,$ again exactly. Every one of those integers vanishes from the index corresponding to $n = 15+1$ onwards, and the run continued until $S + 1$ consecutive zeros had been seen. □

Remark 3. The splitting is what makes the argument finite, and it is exact rather than an over-count only because every array here has a subblock. For a one-column or one-row shape there is no subblock at all, every value of c would be vacuously admissible, and the sum would count each array $12 + 1$ times; those shapes are excluded by $L \ge 2$ and $4 \ge 2$.

4 Verification

Three checks, each independent of the algebra above.

First, the right-hand side of (2) was evaluated at every one of the 22 published indices and agrees with the entry's DATA exactly, with no shift fitted. A double count in the splitting, or a wrong bound on c, would show up here immediately.

Second, the recurrence (1) was evaluated on those published terms in exact integer arithmetic, with no matrices involved, and vanishes wherever it applies.

Third, the annihilation test of Lemma 1 was carried to the full Cayley–Hamilton bound in exact integer arithmetic rather than to a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A184023.

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[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.